

Software Modeling and UML

Experiment 5

State Machine Diagram Modeling

Objectives

- Be familiar with the use of general elements of state machine diagram
- Be familiar with the general process of modeling state machine diagram

Case one: Book

- Consider the several states of a book in the Library management system, draw the state machine diagram to indicate the transition of states for a book
- Different transitions may occur between states
- You can define any reasonable details

Case two: Human

- Consider the several states of a human in a Human resource management system, draw the state machine diagram to indicate the transition of states for a human
- Various states of a human will be different depending on a specific company
 - In general, the states of a human may include
 - On active duty
 - On leave
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- The above description is for reference only.
- You can define any reasonable details

Case three: Research Project

- Build a state machine diagram for a research project in a research project management system
- Draw the state machine diagram
- Briefly explain the states and transitions
- You can define any reasonable details

Please consider the following questions

- Whether you can express composite state?
- Whether you can express entry event and exit event?
- Whether you can express the internal transition, and if the internal transition will activate the entry event?

Experiment requirements

- Please send your work through the link provided by the teaching assistant's cloud storage.
- Requirements for the report
 - Report name: Student ID Name UML EX5 State Machine diagram
 - Please do not compress any document (不要压缩)
 - The deadline for task submission is 6 PM this Saturday evening
- Make sure to send the report following the specified guidelines; otherwise, we will not be able to receive your submission.